



**P1 AURORA GRID**  
**RTU Modbus Specification.**  
(3I56001F100G)  
(Related Firmware Version A004)

January 2013, Version 1. Rev 8

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## THE MODBUS PROTOCOL

In the our Modbus Interface is implemented only the RTU Mode.

The parameters are referred to as 16-bit Modbus Registers, a 32-bit Float parameters covers therefore two Modbus registers

The broadcast mode is not allowed.

An implemented subset of the Modbus functions contains:

- 03 Read-Write Holdings Registers
- 04 Read Input Register

The communication foresees the following setting:

- RS 485
- Baud Rate 19200 bps
- 1 bit start
- 1 bit stop
- No parity
- 8 data bits (RTU protocol)

The Modbus Interface collects data by max 32 Aurora Grid Inverters

The Modbus Functionality was tested with Modscan32 (Version 7.A00-04) program.

## READ OUTPUTS REGISTERS (03)

This function allows to require the value of 16-bit register containing Modbus Address parameter.  
This function allows also to change this parameter, the value changed is saved in EEprom memory.  
The Default parameter is 0x01

### Register Map.

| Add. | Word | Description    | Unit            | Format       |
|------|------|----------------|-----------------|--------------|
| 0001 | 1    | Modbus Address | Decimal (1-247) | Unsigned int |

This function allows to synchronize Modbus collection Data.

### Register Map.

| Add. | Word | Description  | Unit                | Format       |
|------|------|--------------|---------------------|--------------|
| 0002 | 1    | Modbus Reset | Decimal<br>1= Reset | Unsigned int |

## READ INPUT REGISTER (04)

This function allows to require the value of 16-bit register containing the measures of the Modules present in the system .

### Register Map.

| Add. | Word | Description                                      | Unit                                                                         | Format       |
|------|------|--------------------------------------------------|------------------------------------------------------------------------------|--------------|
| 0001 | 1    | Number Modules Aurora Grid Present in the System | Decimal (0-32)                                                               | Unsigned Int |
| 0002 | 1    | Presence Module Aurora Grid 01                   | Decimal<br>(1=Present )<br>(0=Not Present)                                   | Unsigned Int |
| .... | .... | ....                                             | ....                                                                         | ....         |
| 0033 | 1    | Presence Module Aurora Grid 32                   | Decimal<br>(1=Present )<br>(0=Not Present)                                   | Unsigned Int |
| 0034 | 1    | Status Module Aurora Grid01                      | Decimal<br>(0=OK)<br>(1=WARNING)<br>(2= ERROR)<br>(3=COMM.LOST)              |              |
| .... | .... | ....                                             | ....                                                                         | ....         |
| 0065 | 1    | Status Module Aurora Grid32                      | Decimal<br>(0=OK)<br>(1=WARNING)<br>(2= ERROR)<br>(3=COMM.LOST)              | Unsigned Int |
| .... | .... | ....                                             | ....                                                                         | ....         |
| 0098 | 1    | Status Modbus Interface                          | Decimal<br>(0=OK)<br>(1=EEPROM FAILURE)<br>(2= MODULES COMUNICATION FAILURE) | Unsigned Int |

| Add.  | Word | Description                   | Unit                      | Format       |
|-------|------|-------------------------------|---------------------------|--------------|
| 20100 | 1    | Measures Module Aurora Grid01 | View Reg.Map<br>Module 01 | Unsigned Int |
| ....  | .... | ....                          | ....                      | ....         |
| 20121 | 1    | Measures Module Aurora Grid01 | View Reg.Map<br>Module 01 | Unsigned Int |
| 20130 | 2    | Measures Module Aurora Grid01 | View Reg.Map<br>Module 01 | Float        |
| 20132 | 2    | Measures Module Aurora Grid01 | View Reg.Map<br>Module 01 | Float        |
| ....  | .... | ....                          | ....                      | ....         |
| 20168 | 2    | Measures Module Aurora Grid01 | View Reg.Map<br>Module 01 | Float        |

|      |      |      |      |      |
|------|------|------|------|------|
| .... | .... | .... | .... | .... |
|------|------|------|------|------|

| Add.  | Word | Description                   | Unit                      | Format       |
|-------|------|-------------------------------|---------------------------|--------------|
| 23200 | 1    | Measures Module Aurora Grid32 | View Reg.Map<br>Module 01 | Unsigned Int |
| ....  | .... | ....                          | ....                      | ....         |
| 23221 | 1    | Measures Module Aurora Grid32 | View Reg.Map<br>Module 01 | Unsigned Int |
| 23230 | 2    | Measures Module Aurora Grid32 | View Reg.Map<br>Module 01 | Float        |
| 23232 | 2    | Measures Module Aurora Grid32 | View Reg.Map<br>Module 01 | Float        |
| ....  | .... | ....                          | ....                      | ....         |
| 23268 | 2    | Measures Module Aurora Grid32 | View Reg.Map<br>Module 01 | Float        |

**The section below it is only valid if the module belong at these type of Inverters:**

1. **TRIO-27.6-TL-OUTD**
2. **TRIO-20-TL-OUTD**

| Add.  | Word | Description                                 | Unit                                         | Format       |
|-------|------|---------------------------------------------|----------------------------------------------|--------------|
| 24100 | 1    | Measures Aux Board<br>Module Aurora Grid 01 | Register Map<br>Auxiliary<br>Board Module 01 | Unsigned Int |
| ....  | .... | ....                                        | ....                                         | ....         |
| 24145 | 1    | Measures Aux Board<br>Module Aurora Grid 01 | Register Map<br>Auxiliary<br>Board Module 01 | Unsigned Int |
| 24146 | 2    | Measures Aux Board<br>Module Aurora Grid 01 | Register Map<br>Auxiliary<br>Board Module 01 | Float        |
| 24148 | 2    | Measures Aux Board<br>Module Aurora Grid 01 | Register Map<br>Auxiliary<br>Board Module 01 | Float        |
| ....  | .... | ....                                        | ....                                         | ....         |
| 24178 | 2    | Measures Aux Board<br>Module Aurora Grid 01 | Register Map<br>Auxiliary<br>Board Module 01 | Float        |

|      |      |      |      |      |
|------|------|------|------|------|
| .... | .... | .... | .... | .... |
|------|------|------|------|------|

| Add.  | Word | Description                                 | Unit                                         | Format       |
|-------|------|---------------------------------------------|----------------------------------------------|--------------|
| 27200 | 1    | Measures Aux Board<br>Module Aurora Grid 32 | Register Map<br>Auxiliary<br>Board Module 01 | Unsigned Int |
| ....  | .... | ....                                        | ....                                         | ....         |
| 27245 | 1    | Measures Aux Board<br>Module Aurora Grid 32 | Register Map<br>Auxiliary<br>Board Module 01 | Unsigned Int |
| 27246 | 2    | Measures Aux Board<br>Module Aurora Grid 32 | Register Map<br>Auxiliary<br>Board Module 01 | Float        |
| 27248 | 2    | Measures Aux Board<br>Module Aurora Grid 32 | Register Map<br>Auxiliary<br>Board Module 01 | Float        |
| ....  | .... | ....                                        | ....                                         | ....         |
| 27278 | 2    | Measures Aux Board<br>Module Aurora Grid 32 | Register Map<br>Auxiliary<br>Board Module 01 | Float        |

This function allows requiring the value of 16-bit register containing the values concerning the type of equipment in the system.

## Register Map.

| Add. | Word | Description           | Unit                                 | Format       |
|------|------|-----------------------|--------------------------------------|--------------|
| 3500 | 1    | Aurora Type Module 01 | Decimal (ASCII )<br>( See Table 1.1) | Unsigned Int |
| 3501 | 1    | Grid Type Module 01   | Decimal (ASCII )<br>( See Table 1.2) | Unsigned Int |
| 3502 | 1    | Trafo Type Module 01  | Decimal (ASCII )<br>( See Table 1.3) | Unsigned Int |
| 3503 | 1    | Wind Type Module 01   | Decimal (ASCII )<br>( See Table 1.4) | Unsigned Int |

|      |      |      |      |      |
|------|------|------|------|------|
| .... | .... | .... | .... | .... |
|------|------|------|------|------|

| Add. | Word | Description           | Unit                                 | Format       |
|------|------|-----------------------|--------------------------------------|--------------|
| 3624 | 1    | Aurora Type Module 32 | Decimal (ASCII )<br>( See Table 1.1) | Unsigned Int |
| 3625 | 1    | Grid Type Module 32   | Decimal (ASCII )<br>( See Table 1.2) | Unsigned Int |
| 3626 | 1    | Trafo Type Module 32  | Decimal (ASCII )<br>( See Table 1.3) | Unsigned Int |
| 3627 | 1    | Wind Type Module 32   | Decimal (ASCII )<br>( See Table 1.4) | Unsigned Int |

This function allows to require the value of 16-bit register containing the Firmware Revision Modbus Interface

## Register Map.

| Add. | Word | Description            | Unit                                | Format       |
|------|------|------------------------|-------------------------------------|--------------|
| 3650 | 1    | Modbus Interface Type  | Decimal (ASCII )<br>A=Aurora String | Unsigned Int |
| 3651 | 1    | Firmware Major Release | Decimal (ASCII )                    | Unsigned Int |
| 3652 | 1    | Firmware Minor Release | Decimal (ASCII )                    | Unsigned Int |
| 3653 | 1    | Firmware Build Release | Decimal (ASCII )                    | Unsigned Int |



This function allows to require the value of 16-bit register containing the collection of Energy Modules

## Register Map.

| Add. | Word | Description                              | Unit      | Format        |
|------|------|------------------------------------------|-----------|---------------|
| 3700 | 2    | Collection Daily Energy of all Modules   | KWattHour | IEEE 32 Float |
| 3702 | 2    | Collection Total Energy of all Modules   | KWattHour | IEEE 32 Float |
| 3704 | 2    | Collection Partial Energy of all Modules | KWattHour | IEEE 32 Float |
| 3706 | 2    | Collection Week Energy of all Modules    | KWattHour | IEEE 32 Float |
| 3708 | 2    | Collection Monthly Energy of all Modules | KWattHour | IEEE 32 Float |
| 3710 | 2    | Collection Yearly Energy of all Modules  | KWattHour | IEEE 32 Float |
| 3712 | 2    | Collection Input Power of all Modules    | KWatt     | IEEE 32 Float |
| 3714 | 2    | Collection Output Power of all Modules   | KWatt     | IEEE 32 Float |

This function allows requiring the value of 16-bit register containing the values concerning the last 4 Errors.

**These registers will be cleared after being read.**

## Register Map.

| Add. | Word | Description             | Unit                        | Format       |
|------|------|-------------------------|-----------------------------|--------------|
| 3800 | 1    | Last 4 Errors Module 01 | Decimal<br>( See Table 2.4) | Unsigned Int |
| 3801 | 1    | Last 4 Errors Module 01 | Decimal<br>( See Table 2.4) | Unsigned Int |
| 3802 | 1    | Last 4 Errors Module 01 | Decimal<br>( See Table 2.4) | Unsigned Int |
| 3803 | 1    | Last 4 Errors Module 01 | Decimal<br>( See Table 2.4) | Unsigned Int |

|      |      |      |      |      |
|------|------|------|------|------|
| .... | .... | .... | .... | .... |
|------|------|------|------|------|

| Add. | Word | Description             | Unit                        | Format       |
|------|------|-------------------------|-----------------------------|--------------|
| 3924 | 1    | Last 4 Errors Module 32 | Decimal<br>( See Table 2.4) | Unsigned Int |
| 3925 | 1    | Last 4 Errors Module 32 | Decimal<br>( See Table 2.4) | Unsigned Int |
| 3926 | 1    | Last 4 Errors Module 32 | Decimal<br>( See Table 2.4) | Unsigned Int |
| 3927 | 1    | Last 4 Errors Module 32 | Decimal<br>( See Table 2.4) | Unsigned Int |

## Register Map. Module 01.

The below register map is referred at the Module 01 , but it is valid for all the Modules **Aurora Grid** presents in the system.

| Add.  | Word | Description                           | Unit                            | Format        |
|-------|------|---------------------------------------|---------------------------------|---------------|
| 20100 | 1    | Step                                  | Decimal                         | Unsigned Int  |
| 20101 | 1    | SN0 Module 01                         | Decimal (ASCII )<br>(See Note1) | Unsigned Int  |
| ....  | .... | ....                                  | ....                            | ....          |
| 20106 | 1    | SN5 Module 01                         | Decimal (ASCII )<br>(See Note1) | Unsigned Int  |
| 20107 | 1    | PN0 Module 01                         | Decimal (ASCII )<br>(See Note1) | Unsigned Int  |
| ....  | .... | ....                                  | ....                            | ....          |
| 20112 | 1    | PN5 Module 01                         | Decimal (ASCII )<br>(See Note1) | Unsigned Int  |
| 20113 | 1    | Status0 Module 01<br>(Global State)   | Decimal<br>( See Table 2.1)     | Unsigned Int  |
| 20114 | 1    | Status1 Module 01<br>(DcDc State Ch1) | Decimal<br>( See Table 2.3)     | Unsigned Int  |
| 20115 | 1    | Status2 Module 01<br>(DcDc State Ch2) | Decimal<br>( See Table 2.3)     |               |
| 20116 | 1    | Status3 Module 01<br>(Inverter State) | Decimal<br>( See Table 2.2)     | Unsigned Int  |
| 20117 | 1    | Status4 Module 01<br>(Alarm State)    | Decimal<br>( See Table 2.4)     | Unsigned Int  |
| 20118 | 1    | High Week Manufacture                 | Decimal (ASCII )<br>(See Note1) | Unsigned Int  |
| 20119 | 1    | Low Week Manufacture                  | Decimal (ASCII )<br>(See Note1) | Unsigned Int  |
| 20120 | 1    | High Year Manufacture                 | Decimal (ASCII )<br>(See Note1) | Unsigned Int  |
| 20121 | 1    | Low Year Manufacture                  | Decimal (ASCII )<br>(See Note1) | Unsigned Int  |
| ....  | .... | ....                                  | ....                            | ....          |
| 20130 | 2    | Daily Energy Module 01                | WattHour                        | IEEE 32 Float |
| 20132 | 2    | Total Energy Module 01                | KWattHour                       | IEEE 32 Float |
| 20134 | 2    | Partial Energy Module 01              | KWattHour                       | IEEE 32 Float |
| 20136 | 2    | Week Energy Module 01                 | KWattHour                       | IEEE 32 Float |
| 20138 | 2    | Monthly Energy Module 01              | KWattHour                       | IEEE 32 Float |
| 20140 | 2    | Yearly Energy Module 01               | KWattHour                       | IEEE 32 Float |
| 20142 | 2    | 3 phases Grid Voltage Module 01       | Volt Rms                        | IEEE 32 Float |

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|       |   |                                       |                |               |
|-------|---|---------------------------------------|----------------|---------------|
| 20144 | 2 | 3 phases Grid Current Module 01       | Ampere Rms     | IEEE 32 Float |
| 20146 | 2 | 3 phases Grid power Module 01         | Watt           | IEEE 32 Float |
| 20148 | 2 | 3 phases Frequency Module 01          | Hz             | IEEE 32 Float |
| 20150 | 2 | Input 1 Power Module 01               | Watt           | IEEE 32 Float |
| 20152 | 2 | Input1 Voltage Module 01              | Volt           | IEEE 32 Float |
| 20154 | 2 | Input1 Current Module 01              | Ampere         | IEEE 32 Float |
| 20156 | 2 | Input 2 Power Module 01               | Watt           | IEEE 32 Float |
| 20158 | 2 | Input2 Voltage Module 01              | Volt           | IEEE 32 Float |
| 20160 | 2 | Input2 Current Module 01              | Ampere         | IEEE 32 Float |
| 20162 | 2 | Inverter Temperature Module 01        | Celsius Degree | IEEE 32 Float |
| 20164 | 2 | Booster Temperature Module 01         | Celsius Degree | IEEE 32 Float |
| 20166 | 2 | Isolation Resistance Module 01        | Mohm           | IEEE 32 Float |
| 20168 | 2 | Wind generator Frequency<br>Module 01 | Hz             | IEEE 32 Float |

## Register Map Auxiliary Board Module 01

The below register map is referred at the Auxiliary Board Inverter 01, it is valid for all the Auxiliary Board presents in the system.

| Add.  | Word | Description                                 | Unit                                                        | Format       |
|-------|------|---------------------------------------------|-------------------------------------------------------------|--------------|
| 24100 | 1    | Presence Auxiliary Board Module 01          | Decimal<br>0= Aux Board not present<br>1= Aux Board present | Unsigned Int |
| 24101 | 1    | SN0 Inverter Module 01                      | Decimal (ASCII )<br>( See Note1)                            | Unsigned Int |
| ....  | .... | ....                                        | ....                                                        | ....         |
| 24106 | 1    | SN5 Inverter Module 01                      | Decimal (ASCII )<br>( See Note1)                            | Unsigned Int |
| 24107 | 1    | SN0 Auxiliary Communication Board Module 01 | Decimal (ASCII )<br>( See Note1)                            | Unsigned Int |
| ....  | .... | ....                                        | ....                                                        | ....         |
| 24112 | 1    | SN5 Auxiliary Communication Board Module 01 | Decimal (ASCII )<br>( See Note1)                            | Unsigned Int |
| 24113 | 1    | PN0 Auxiliary Board Communication           | Decimal (ASCII )<br>( See Note1)                            | Unsigned Int |
| ....  | .... | ....                                        | ....                                                        | ....         |
| 24118 | 1    | PN5 Auxiliary Communication Board           | Decimal (ASCII )<br>( See Note1)                            | Unsigned Int |
| 24119 | 1    | Week Production (High) Communication Board  | Decimal (ASCII )<br>( See Note1)                            | Unsigned Int |
| 24120 | 1    | Week Production(Low) Communication Board    | Decimal (ASCII )                                            | Unsigned Int |
| 24121 | 1    | Year Production (High) Communication Board  | Decimal (ASCII )<br>( See Note1)                            | Unsigned Int |
| 24122 | 1    | Year Production(Low) Communication Board    | Decimal (ASCII )                                            | Unsigned Int |
| 24123 | 1    | SN0 Auxiliary Fuse Control Board            | Decimal (ASCII )<br>( See Note1)                            | Unsigned Int |
| ....  | .... | ....                                        | ....                                                        | ....         |
| 24128 | 1    | SN5 Auxiliary Fuse Control Board            | Decimal (ASCII )<br>( See Note1)                            | Unsigned Int |
| 24129 | 1    | PN0 Auxiliary Fuse Control Board            | Decimal (ASCII )<br>( See Note1)                            | Unsigned Int |
| ....  | .... | ....                                        | ....                                                        | ....         |
| 24134 | 1    | PN5 Auxiliary Fuse Control Board            | Decimal (ASCII )<br>( See Note1)                            | Unsigned Int |
| 24135 | 1    | Week Production (High)                      | Decimal (ASCII )                                            | Unsigned Int |

|       |   |                                              |                                 |               |
|-------|---|----------------------------------------------|---------------------------------|---------------|
|       |   | Fuse Control Board                           | ( See Note1)                    |               |
| 24136 | 1 | Week Production(Low)<br>Fuse Control Board   | Decimal (ASCII)                 | Unsigned Int  |
| 24137 | 1 | Year Production (High)<br>Fuse Control Board | Decimal (ASCII)<br>( See Note1) | Unsigned Int  |
| 24138 | 1 | Year Production(Low)<br>Fuse Control Board   | Decimal (ASCII)                 | Unsigned Int  |
| 24139 | 1 | State Communication Board<br>Byte 4          | Decimal<br>(See Table 3.1)      | Unsigned Int  |
| 24140 | 1 | State Communication Board<br>Byte 5          | Decimal<br>(See Table 3.1)      | Unsigned Int  |
| 24141 | 1 | State Fuse Control Board Byte 0              | Decimal<br>(See Table 4.1)      | Unsigned Int  |
| 24142 | 1 | State Fuse Control Board Byte 1              | Decimal<br>(See Table 4.1)      | Unsigned Int  |
| 24143 | 1 | State Fuse Control Board Byte 2              | Decimal<br>(See Table 4.1)      | Unsigned Int  |
| 24144 | 1 | State Fuse Control Board Byte 3              | Decimal<br>(See Table 4.1)      | Unsigned Int  |
| 24145 | 1 | State Fuse Control Board Byte 4              | Decimal<br>(See Table 4.1)      | Unsigned Int  |
| 24146 | 2 | PT100 Communication Board                    | Celsius Degree                  | IEEE 32 Float |
| 24148 | 2 | PT1000 Communication Board                   | Celsius Degree                  | IEEE 32 Float |
| 24150 | 2 | Analog Input 1 Communication<br>Board        | N/A                             | IEEE 32 Float |
| 24152 | 2 | Analog Input 2 Communication<br>Board        | N/A                             | IEEE 32 Float |
| 24154 | 2 | NTC Communication Board                      | Celsius Degree                  | IEEE 32 Float |
| 24156 | 2 | Voltage Ch1 Fuse Control Board               | Volt                            | IEEE 32 Float |
| 24158 | 2 | Voltage Ch2 Fuse Control Board               | Volt                            | IEEE 32 Float |
| 24160 | 2 | Current 1A Fuse Control Board                | Ampere                          | IEEE 32 Float |
| 24162 | 2 | Current 1B Fuse Control Board                | Ampere                          | IEEE 32 Float |
| 24164 | 2 | Current 1C Fuse Control Board                | Ampere                          | IEEE 32 Float |
| 24166 | 2 | Current 1D Fuse Control Board                | Ampere                          | IEEE 32 Float |
| 24168 | 2 | Current 1E Fuse Control Board                | Ampere                          | IEEE 32 Float |
| 24170 | 2 | Current 2A Fuse Control Board                | Ampere                          | IEEE 32 Float |
| 24172 | 2 | Current 2B Fuse Control Board                | Ampere                          | IEEE 32 Float |
| 24174 | 2 | Current 2C Fuse Control Board                | Ampere                          | IEEE 32 Float |
| 24176 | 2 | Current 2D Fuse Control Board                | Ampere                          | IEEE 32 Float |
| 24178 | 2 | Current 2E Fuse Control Board                | Ampere                          | IEEE 32 Float |

**Table 1.1 - Model identifier. See table below**

| Model Char |                | Indoor/Outdoor and type                               |
|------------|----------------|-------------------------------------------------------|
| 105        | (ASCII -> 'i') | PVI-2000-IND                                          |
| 111        | (ASCII -> 'o') | PVI-2000-OUTD                                         |
| 108        | (ASCII -> 'l') | PVI-3600-IND                                          |
| 79         | (ASCII -> 'O') | PVI-3600-OUTD                                         |
| 53         | (ASCII -> '5') | PVI-5000-OUTD                                         |
| 54         | (ASCII -> '6') | PVI-6000-OUTD                                         |
| 80         | (ASCII -> 'P') | 3-phase interface (3G74)                              |
| 67         | (ASCII -> 'C') | PVI-CENTRAL-50 module                                 |
| 52         | (ASCII -> '4') | PVI-4.2-OUTD                                          |
| 51         | (ASCII -> '3') | PVI-3.6-OUTD                                          |
| 50         | (ASCII -> '2') | PVI-3.3-OUTD                                          |
| 49         | (ASCII -> '1') | PVI-3.0-OUTD                                          |
| 68         | (ASCII -> 'D') | PVI-12.5-OUTD                                         |
| 88         | (ASCII -> 'X') | PVI-10.0-OUTD                                         |
| 72         | (ASCII -> 'H') | PVI-4.6-OUTD-I                                        |
| 104        | (ASCII -> 'h') | PVI-3.8-OUTD-I                                        |
| 84         | (ASCII -> 'T') | PVI-12.5-OUTD-I (output 480 VAC)                      |
| 116        | (ASCII -> 't') | PVI-10.0-OUTD-I (output 480 VAC)                      |
| 85         | (ASCII -> 'U') | PVI-12.5-OUTD-I (output 208 VAC)                      |
| 117        | (ASCII -> 'u') | PVI-10.0-OUTD-I (output 208 VAC)                      |
| 86         | (ASCII -> 'V') | PVI-12.5-OUTD-I (output 380 VAC)                      |
| 118        | (ASCII -> 'v') | PVI-10.0-OUTD-I (output 380 VAC)                      |
| 90         | (ASCII -> 'Z') | PVI-12.5-OUTD-I (output 600 VAC)                      |
| 122        | (ASCII -> 'z') | PVI-10.0-OUTD-I (output 600 VAC)                      |
| 77         | (ASCII -> 'M') | PVI-CENTRAL-250                                       |
| 119        | (ASCII -> 'w') | PVI-10.0-I-OUTD (output 480 VAC – current limit 12 A) |
| 89         | (ASCII -> 'Y') | TRIO-27.6-TL-OUTD (output 400 VAC)                    |
| 121        | (ASCII -> 'y') | TRIO-20-TL-OUTD (output 400 VAC)                      |
| 103        | (ASCII -> 'g') | UNO-2.0-OUTD                                          |
| 71         | (ASCII -> 'G') | UNO-2.5-OUTD                                          |
| 76         | (ASCII -> 'L') | PVI-CENTRAL-350 Liquid Cooled (control board)         |
| 66         | (ASCII -> 'B') | PVI-CENTRAL-350 Liquid Cooled (display board)         |
| 65         | (ASCII -> 'A') | PVI-CENTRAL-350 Liquid Cooled (AC gathering)          |
| 69         | (ASCII -> 'E') | PVI-CENTRAL-67 module                                 |
| 99         | (ASCII -> 'c') | CDD                                                   |
| 70         | (ASCII -> 'F') | TRIO-27.6-TL-OUTD (output 480 VAC)                    |
| 102        | (ASCII -> 'f') | TRIO-20-TL-OUTD (output 480 VAC)                      |
| 78         | (ASCII -> 'N') | PVI-12.5-OUTD Universal (output 400 VAC)              |
| 110        | (ASCII -> 'n') | PVI-10.0-OUTD Universal (output 400 VAC)              |

|     |                |                                                 |
|-----|----------------|-------------------------------------------------|
| 81  | (ASCII -> 'Q') | PVI-8.0-OUTD Universal PLUS<br>(output 400 VAC) |
| 113 | (ASCII -> 'q') | PVI-8.0-OUTD Universal (output 400 VAC)         |
| 74  | (ASCII -> 'J') | PVI-1700-OUTD                                   |
| 106 | (ASCII -> 'j') | PVI-1700-IND                                    |
|     |                |                                                 |



**Table 1.2 - Grid-Standard identifier. See table below.**

| Grid-standard char |                | Grid standard                           |
|--------------------|----------------|-----------------------------------------|
| 65                 | (ASCII -> 'A') | USA - UL1741                            |
| 69                 | (ASCII -> 'E') | Germany - VDE0126                       |
| 83                 | (ASCII -> 'S') | Spain - DR 1663/2000                    |
| 73                 | (ASCII -> 'I') | Italy - ENEL DK 5950                    |
| 85                 | (ASCII -> 'U') | UK - UK G83                             |
| 75                 | (ASCII -> 'K') | Australia - AS 4777                     |
| 70                 | (ASCII -> 'F') | France - VDE French Model               |
| 82                 | (ASCII -> 'R') | Ireland - EN50438                       |
| 66                 | (ASCII -> 'B') | Belgium - VDE Belgium Model             |
| 79                 | (ASCII -> 'O') | Korea                                   |
| 71                 | (ASCII -> 'G') | Greece - VDE Greece Model               |
| 84                 | (ASCII -> 'T') | Taiwan                                  |
| 67                 | (ASCII -> 'C') | Czech Republic                          |
| 81                 | (ASCII -> 'Q') | People's Republic of China              |
| 63                 | (ASCII -> '?') | China                                   |
| 97                 | (ASCII -> 'a') | USA - UL1741<br>Vout = 208 single phase |
| 98                 | (ASCII -> 'b') | USA - UL1741<br>Vout = 240 single phase |
| 99                 | (ASCII -> 'c') | USA - UL1741<br>Vout = 277 single phase |
| 88                 | (ASCII -> 'X') | Debug Standard 1                        |
| 120                | (ASCII -> 'x') | Debug Standard 2                        |
| 117                | (ASCII -> 'u') | UK – UK-G59                             |
| 107                | (ASCII -> 'k') | Israel – Derived from AS 4777           |
| 87                 | (ASCII -> 'W') | Germany – BDEW                          |
| 72                 | (ASCII -> 'H') | Hungary                                 |
| 111                | (ASCII -> 'o') | Corsica                                 |
| 80                 | (ASCII -> 'P') | Portugal                                |
| 101                | (ASCII -> 'e') | VDE AR-N-4105                           |
| 105                | (ASCII -> 'i') | CEI 021 ( internal protection)          |
| 106                | (ASCII -> 'j') | CEI 021 ( external protection)          |
| 68                 | (ASCII -> 'D') | UL for Canada                           |
| 102                | (ASCII -> 'f') | EDF for France                          |
| 113                | (ASCII -> 'q') | Australia Queensland                    |
| 115                | (ASCII -> 's') | Slovenia                                |
| 114                | (ASCII -> 'r') | South Africa                            |
| 112                | (ASCII -> 'p') | SPAIN RD 1565                           |

|     |                |                  |
|-----|----------------|------------------|
| 109 | (ASCII -> 'm') | BELG C10-11 100% |
| 77  | (ASCII -> 'M') | BELG C10-11 110% |
| 90  | (ASCII -> 'Z') | BRAZIL           |

**Table 1.3 - Transformer/Transformerless identifier. See table below**

| <b>Transformer char</b> |                | <b>Transformer/Transformerless version</b> |
|-------------------------|----------------|--------------------------------------------|
| 78                      | (ASCII -> 'N') | Transformerless version                    |
| 84                      | (ASCII -> 'T') | Transformer version                        |
| 116                     | (ASCII -> 't') | Transformer HF version                     |
| 120                     | (ASCII -> 'x') | Dummy transformer type                     |

**Table 1.4 - Photovoltaic/eolic identifier. See table below**

| <b>Type char</b> |                | <b>Photovoltaic/eolic version</b> |
|------------------|----------------|-----------------------------------|
| 80               | (ASCII -> 'P') | Photovoltaic version              |
| 87               | (ASCII -> 'W') | Eolic version                     |
| 120              | (ASCII -> 'x') | Dummy inverter type               |

**Table2.1 Status 0(Global State). See table below**

| <b>Global State</b> |                          |
|---------------------|--------------------------|
| <b>0</b>            | Sending Parameters       |
| <b>1</b>            | Wait Sun/Grid            |
| <b>2</b>            | Checking Grid            |
| <b>3</b>            | Measuring Riso           |
| <b>4</b>            | DcDc Start               |
| <b>5</b>            | Inverter Start           |
| <b>6</b>            | Run                      |
| <b>7</b>            | Recovery                 |
| <b>8</b>            | Pause                    |
| <b>9</b>            | Ground Fault             |
| <b>10</b>           | OTH Fault                |
| <b>11</b>           | Address Setting          |
| <b>12</b>           | Self Test                |
| <b>13</b>           | Self Test Fail           |
| <b>14</b>           | Sensor Test + Meas.Riso  |
| <b>15</b>           | Leak Fault               |
| <b>16</b>           | Waiting for manual reset |
| <b>17</b>           | Internal Error E026      |
| <b>18</b>           | Internal Error E027      |
| <b>19</b>           | Internal Error E028      |
| <b>20</b>           | Internal Error E029      |
| <b>21</b>           | Internal Error E030      |
| <b>22</b>           | Sending Wind Table       |
| <b>23</b>           | Failed Sending table     |
| <b>24</b>           | UTH Fault                |
| <b>25</b>           | Remote OFF               |
| <b>26</b>           | Interlock Fail           |
| <b>27</b>           | Executing Autotest       |
| <b>30</b>           | Waiting Sun              |
| <b>31</b>           | Temperature Fault        |
| <b>32</b>           | Fan Stauked              |
| <b>33</b>           | Int. Com. Fault          |
| <b>34</b>           | Slave Insertion          |
| <b>35</b>           | DC Switch Open           |
| <b>36</b>           | TRAS Switch Open         |
| <b>37</b>           | MASTER Exclusion         |
| <b>38</b>           | Auto Exclusion           |
| <b>98</b>           | Erasing Internal EEprom  |
| <b>99</b>           | Erasing External EEprom  |
| <b>100</b>          | Counting EEprom          |
| <b>101</b>          | Freeze                   |
| <b>200</b>          | Dsp Programming          |

**Table2.2 Status 1 (Inverter State). See table below**

| <b>Inverter State</b> |                                    |
|-----------------------|------------------------------------|
| <b>0</b>              | Stand By                           |
| <b>1</b>              | Checking Grid                      |
| <b>2</b>              | Run                                |
| <b>3</b>              | Bulk OV                            |
| <b>4</b>              | Out OC                             |
| <b>5</b>              | IGBT Sat                           |
| <b>6</b>              | Bulk UV                            |
| <b>7</b>              | Degauss Error                      |
| <b>8</b>              | No Parameters                      |
| <b>9</b>              | Bulk Low                           |
| <b>10</b>             | Grid OV                            |
| <b>11</b>             | Communication Error                |
| <b>12</b>             | Degaussing                         |
| <b>13</b>             | Starting                           |
| <b>14</b>             | Bulk Cap Fail                      |
| <b>15</b>             | Leak Fail                          |
| <b>16</b>             | DcDc Fail                          |
| <b>17</b>             | Ileak Sensor Fail                  |
| <b>18</b>             | SelfTest: relay inverter           |
| <b>19</b>             | SelfTest: wait for sensor test     |
| <b>20</b>             | SelfTest: test relay DcDc + sensor |
| <b>21</b>             | SelfTest: relay inverter fail      |
| <b>22</b>             | SelfTest timeout fail              |
| <b>23</b>             | SelfTest: relay DcDc fail          |
| <b>24</b>             | Self Test 1                        |
| <b>25</b>             | Waiting self test start            |
| <b>26</b>             | Dc Injection                       |
| <b>27</b>             | Self Test 2                        |
| <b>28</b>             | Self Test 3                        |
| <b>29</b>             | Self Test 4                        |
| <b>30</b>             | Internal Error                     |
| <b>31</b>             | Internal Error                     |
| <b>40</b>             | Forbidden State                    |
| <b>41</b>             | Input UC                           |
| <b>42</b>             | Zero Power                         |
| <b>43</b>             | Grid Not Present                   |
| <b>44</b>             | Waiting Start                      |
| <b>45</b>             | MPPT                               |
| <b>46</b>             | Grid Fail                          |
| <b>47</b>             | Input OC                           |
| <b>...</b>            | ...                                |
| <b>255</b>            | Inverter Dsp not programmed        |

**Table2.3 Status 2 (DcRc State). See table below**

| <b>DcDc State</b> |                         |
|-------------------|-------------------------|
| <b>0</b>          | DcDc OFF                |
| <b>1</b>          | Ramp Start              |
| <b>2</b>          | MPPT                    |
| <b>3</b>          | <i>Not Used</i>         |
| <b>4</b>          | Input OC                |
| <b>5</b>          | Input UV                |
| <b>6</b>          | Input OV                |
| <b>7</b>          | Input Low               |
| <b>8</b>          | No Parameters           |
| <b>9</b>          | Bulk OV                 |
| <b>10</b>         | Communication Error     |
| <b>11</b>         | Ramp Fail               |
| <b>12</b>         | Internal Error          |
| <b>13</b>         | Input mode Error        |
| <b>14</b>         | Ground Fault            |
| <b>15</b>         | Inverter Fail           |
| <b>16</b>         | DcDc IGBT Sat           |
| <b>17</b>         | DcDc ILEAK Fail         |
| <b>18</b>         | DcDc Grid Fail          |
| <b>19</b>         | DcDc Comm. Error        |
| ...               | ...                     |
| <b>255</b>        | DcDc Dsp not programmed |

**Table2.4 Status 3 (Alarm State). See table below**

| <b>Alarm State</b> |                   |      |       |                  |      |
|--------------------|-------------------|------|-------|------------------|------|
| Value              | Description       | Code | Value | Description      | Code |
| 0                  | No Alarm          |      | 34    | Grid OF          | W006 |
| 1                  | Sun Low           | W001 | 35    | Grid UF          | W007 |
| 2                  | Input OC          | E001 | 36    | Z grid Hi        | W008 |
| 3                  | Input UV          | W002 | 37    | Internal error   | E024 |
| 4                  | Input OV          | E002 | 38    | Riso Low         | E025 |
| 5                  | Sun Low           | W001 | 39    | Vref Error       | E026 |
| 6                  | No Parameters     | E003 | 40    | Error Meas V     | E027 |
| 7                  | Bulk OV           | E004 | 41    | Error Meas F     | E028 |
| 8                  | Comm.Error        | E005 | 42    | Error Meas Z     | E029 |
| 9                  | Output OC         | E006 | 43    | Error Meas Ileak | E030 |
| 10                 | IGBT Sat          | E007 | 44    | Error Read V     | E031 |
| 11                 | Bulk UV           | W011 | 45    | Error Read I     | E032 |
| 12                 | Internal error    | E009 | 46    | Table fail       | W009 |
| 13                 | Grid Fail         | W003 | 47    | Fan Fail         | W010 |
| 14                 | Bulk Low          | E010 | 48    | UTH              | E033 |
| 15                 | Ramp Fail         | E011 | 49    | Interlock fail   | E034 |
| 16                 | Dc/Dc Fail        | E012 | 50    | Remote Off       | E035 |
| 17                 | Wrong Mode        | E013 | 51    | Vout Avg error   | E036 |
| 18                 | Ground Fault      | ---  | 52    | Battery low      | W012 |
| 19                 | Over Temp.        | E014 | 53    | Clk fail         | W013 |
| 20                 | Bulk Cap Fail     | E015 | 54    | Input UC         | E037 |
| 21                 | Inverter Fail     | E016 | 55    | Zero Power       | W014 |
| 22                 | Start Timeout     | E017 | 56    | Fan Stuck        | E038 |
| 23                 | Ground Fault      | E018 | 57    | DC Switch Open   | E039 |
| 24                 | Degauss error     | ---  | 58    | Tras Switch Open | E040 |
| 25                 | Ileak sens.fail   | E019 | 59    | AC Switch Open   | E041 |
| 26                 | DcDc Fail         | E012 | 60    | Bulk UV          | E042 |
| 27                 | Self Test Error 1 | E020 | 61    | Autoexclusion    | E043 |
| 28                 | Self Test Error 2 | E021 | 62    | Grid df/dt       | W015 |
| 29                 | Self Test Error 3 | E019 | 63    | Den switch Open  | W016 |
| 30                 | Self Test Error 4 | E022 | 64    | Jbox fail        | W017 |
| 31                 | DC inj error      | E023 | 65    | DC Door Open     | E044 |
| 32                 | Grid OV           | W004 | 66    | AC Door Open     | E045 |
| 33                 | Grid UV           | W005 |       |                  |      |

**Table3.1 Communication Board States. See table below**

|               | <b>Communication Board</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Byte 0</b> | NONE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Byte 1</b> | NONE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Byte 2</b> | NONE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Byte 3</b> | NONE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Byte 4</b> | - Bit 0 : (0 = OTH not active – OTH is active)*<br>- Bit 1 : NONE<br>...<br>- Bit 7 : NONE                                                                                                                                                                                                                                                                                                                                                                                                                        |
| <b>Byte 5</b> | - Bit 0 : (0 = SD card is absent – 1 = SD card is present)*<br>- Bit 1 : (0 = SD card fault – 1 = SD card is ok)*<br>- Bit 2 : (0 = Exp. board is absent – 1 = Exp. board is present)*<br>- Bit 3 : (0 = Wi-Fi board is absent – 1 = Wi-Fi board is present)*<br>- Bit 4 : (0 = PMU interface is absent – 1 = PMU interface is present)*<br>- Bit 5 : (0 = SPD DC 1 fault – 1 = SPD DC 1 card is ok)*<br>- Bit 6 : (0 = SPD DC 2 fault – 1 = SPD DC 2 is ok)*<br>- Bit 7 : (0 = SPD AC fault – 1 = SPD AC is ok)* |



**Table4.1 Fuse Control Board States. See table below**

| <b>Fuse Control Board</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Byte 0</b>             | <ul style="list-style-type: none"> <li>- Bit 0 : (0 = All strings are ok – 1 = One or more strings are in fault)*</li> <li>- Bit 1 : (0 = Overvoltage not active – 1 = Overvoltage is active)*</li> <li>- Bit 2 : (0 = All currents are balanced – 1 = One or more unbalanced currents)*</li> <li>- Bit 3 : (0 = Overcurrent not active – 1 = Overcurrent is active)*</li> <li>- Bit 4 : (0 = Calibration is ok – 1 = Calibration not ok)*</li> <li>...</li> <li>- Bit 7 : (0 = String selftest passed – 1 = String selftest not passed)*</li> </ul>                                                                                                     |
| <b>Byte 1</b>             | <ul style="list-style-type: none"> <li>- Bit 0 : (0 = String 2C ok – 1 = String 2C fault)*</li> <li>- Bit 1 : (0 = String 2B ok – 1 = String 2B fault)*</li> <li>- Bit 2 : (0 = String 2A ok – 1 = String 2A fault)*</li> <li>- Bit 3 : (0 = String 1E ok – 1 = String 1E fault)*</li> <li>- Bit 4 : (0 = String 1D ok – 1 = String 1D fault)*</li> <li>- Bit 5 : (0 = String 1C ok – 1 = String 1C fault)*</li> <li>- Bit 6 : (0 = String 1B ok – 1 = String 1B fault)*</li> <li>- Bit 7 : (0 = String 1A ok – 1 = String 1A fault)*</li> </ul>                                                                                                         |
| <b>Byte 2</b>             | <ul style="list-style-type: none"> <li>...</li> <li>- Bit 6 : (0 = String 2E ok – 1 = String 2E fault)*</li> <li>- Bit 7 : (0 = String 2D ok – 1 = String 2D fault)*</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <b>Byte 3</b>             | <ul style="list-style-type: none"> <li>- Bit 0 : (0 = Current 2C balanced – 1 = Current 2C unbalanced)*</li> <li>- Bit 1 : (0 = Current 2B balanced – 1 = Current 2B unbalanced)*</li> <li>- Bit 2 : (0 = Current 2A balanced – 1 = Current 2A unbalanced)*</li> <li>- Bit 3 : (0 = Current 1E balanced – 1 = Current 1E unbalanced)*</li> <li>- Bit 4 : (0 = Current 1D balanced – 1 = Current 1D unbalanced)*</li> <li>- Bit 5 : (0 = Current 1C balanced – 1 = Current 1C unbalanced)*</li> <li>- Bit 6 : (0 = Current 1B balanced – 1 = Current 1B unbalanced)*</li> <li>- Bit 7 : (0 = Current 1A balanced – 1 = Current 1A unbalanced)*</li> </ul> |
| <b>Byte 4</b>             | <ul style="list-style-type: none"> <li>...</li> <li>- Bit 6 : (0 = Current 2E balanced – 1 = Current 2E unbalanced)*</li> <li>- Bit 7 : (0 = Current 2D balanced – 1 = Current 2D unbalanced)*</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Byte 5</b>             | NONE                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

## Note 1

### Example Reading Sn:

|      |    |           |
|------|----|-----------|
| Sn 0 | 48 | ASCII '0' |
| Sn 1 | 48 | ASCII '0' |
| Sn 2 | 48 | ASCII '0' |
| Sn 3 | 48 | ASCII '0' |
| Sn 4 | 48 | ASCII '0' |
| Sn 5 | 49 | ASCII '1' |

### Example Reading Pn:

|      |    |           |
|------|----|-----------|
| Pn 0 | 45 | ASCII '-' |
| Pn 1 | 51 | ASCII '3' |
| Pn 2 | 76 | ASCII 'L' |
| Pn 3 | 54 | ASCII '6' |
| Pn 4 | 56 | ASCII '8' |
| Pn 5 | 45 | ASCII '-' |

### Example Reading Week Production:

|      |    |           |
|------|----|-----------|
| Wk 0 | 50 | ASCII '2' |
| Wk 1 | 53 | ASCII '5' |

### Example Reading Year Production:

|      |    |           |
|------|----|-----------|
| Yr 0 | 48 | ASCII '0' |
| Yr 1 | 57 | ASCII '9' |